

## Forensics Incident Detector

### Introduction

The purpose of this tool is to automate the process of analyzing system logs to detect suspicious activities. Digital Forensics often requires going through a large amount of logs and data to find indicators that breach or compromise has happen. This tool streamlines that process, saving investigators time and reducing the likelihood of human error. By automating this process our lightweight tool quickly extracts metadata, suspicious patterns, and generates a readable report that can be used in investigations.

### Technical Information

#### Languages & Libraries:

- Python 3.11 – Chosen for its extensive support for file parsing and data analysis.
- pandas – For efficient log data processing and manipulation.
- re (Regular Expressions) – To extract specific patterns from logs.
- argparse – For command-line argument handling.
- matplotlib – For optional visualization of log activity trends.

#### Techniques Used:

- Log Parsing: The tool reads multiple log formats (e.g., system logs, Apache server logs).
- Pattern Matching: Suspicious patterns such as repeated failed login attempts or access from unusual IP addresses are detected using regex.
- Report Generation: Outputs results in a structured text file or CSV format for review.
- Optional Visualization: Activity over time can be visualized with a graph for easier interpretation.

These were choosing for easy readability and extensive libraries that simplify the handling of all this data which finds pattern recognition.

Results:

First running hydra to brute force to generate an alert

```
—(kali@kali)-[~/forensics]
-$ hydra -l kali -P pass.txt ssh://10.0.2.15
hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use
hydra for military or secret service organizations, or for illegal purposes (this
is in violation of the law in your country. Please read n-binding, these *** ignore laws and ethics anyway).

hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-12-04 00:48
[WARNING] Many SSH configurations limit the number of parallel tasks, it is
recommended to reduce the tasks: use -t 4
[DATA] max 13 tasks per 1 server, overall 13 tasks, 13 login tries (1:1/13)
~1 try per task
[DATA] attacking ssh://10.0.2.15:22/
22][ssh] host: 10.0.2.15 login: kali password: ab12cd34
of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 1 final worker threads did not complete
until end.
[ERROR] 1 target did not resolve or could not be connected
[ERROR] 0 target did not complete
hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-12-04 00:53
```

Next running our tool incidetector.py

Example of HTML report that was generated after attempted brute force

### Incident Detection Report

Collected At: 2025-12-04T01:01:56.661498+00:00

Incident Types: 8

Total Alerts: 139

#### Alerts

| Type        | Details | Timestamp |
|-------------|---------|-----------|
| svc_started | 6       | 70        |
| auth_fail   | 8       | 21        |
| failed_ssh  | 8       | 21        |
| adduser     | 9       | 18        |
| cron_cmd    | 5       | 3         |
| usb_connect | 4       | 3         |
| useradd     | 9       | 1         |
| success_ssh | 3       | 2         |

#### Log Excerpts

```
journalctl.txt:Dec 04 00:37:47 localhost.localdomain kernel: usb usb1: New USB device found, idVendor=1d6b, idProduct=0002, bcdDevice= 6.12
journalctl.txt:Dec 04 00:37:47 localhost.localdomain kernel: usb usb2: New USB device found, idVendor=1d6b, idProduct=0001, bcdDevice= 6.12
journalctl.txt:Dec 04 00:37:47 localhost.localdomain kernel: usb 2-1: New USB device found, idVendor=80ee, idProduct=8023, bcdDevice= 1.00
journalctl.txt:Dec 04 00:37:47 localhost.localdomain systemd[1]: Started systemd-ask-password-wall.path - Forward Password Requests to Wall Directory Watch.
journalctl.txt:Dec 04 00:37:47 localhost.localdomain systemd[1]: Started systemd-journald.service - Journal Service.
journalctl.txt:Dec 04 00:37:47 localhost.localdomain systemd[1]: Started systemd-udevd.service - Rule-based Manager for Device Events and Files.
journalctl.txt:Dec 04 00:37:47 localhost.localdomain systemd[1]: Started plymouth-start.service - Show Plymouth Boot Screen.
```

.JSON output of failed attempts

```
{
  "rule_id": "failed_ssh",
  "message": "Failed SSH login",
  "severity": 8,
  "count": 21,
  "examples": [
    {

```

.Json Shows File Hashes

```
"file_hashes": [  
  {  
    "path": "/etc/passwd",  
    "sha256": "e03757dff8119c32bce98932a9a24b23e681cbf2ee0a1d191b77418eab777c01"  
  },  
  {  
    "path": "/etc/shadow",  
    "sha256": "feafc802f09b6691e4e7ef722d9351b07643880735ca6ea87e971ce925b625db"  
  },  
  {  
    "path": "/etc/hosts",  
    "sha256": "c8673aa850c7ccb0c20985eb03977203a6a139ce755f1b082d70016aac094c36"  
  },  
  {  
    "path": "/etc/ssh/sshd_config",  
    "sha256": "f5c9addcb8a12e271a0e0a8dfe05eda55e1b2c5493b13c5e7ee27a5db8bd42c1"  
  }  
]
```

## Lessons learned & Conclusion

Handling all the multiple log paths was difficult but creating a parser that can handle the different file types helped a ton. Also proper testing with both large and small incidents. What worked the best was getting the automation to generate an output as html and Json format, which saves time compared to doing this manually. What didn't work was the tool couldn't handle the longer logs full of data but we improved it by processing the logs into large chunks, such as using -7 days for past seven days of activity instead of the whole log, using -full instead. I wanted to implement a real time monitoring system rather than more of a post analysis, which would provide real time alerts, that could be the next step.

In conclusion this project was very fun and interesting to work on, I wanted to do something with checking files and hashes but this takes it to another level, from checking logs to outputting the findings using pattern recognition is way beyond cool. This project shows the practical application of programming and data analysis and related back to my major cybersecurity is very relevant in today's age. Project was very cool to work on, in the future this provides a foundation to make a more advanced digital forensics analysis tool.